



Modelling and optimization of natural deep eutectic solvent-assisted extraction for carotenoids targeting lutein for developing food additives from used marigold flowers

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ABSTRACT

Natural deep eutectic solvents (NADES) have proved their potential as a green technology for extracting phytochemicals in recent times. Different NADES combinations with various molar ratios of choline chloride as a hydrogen bond acceptor (HBA) and glucose/tartaric acid as a hydrogen bond donor (HBD) were screened for better extraction efficiency of carotenoids from a rich source-dried marigold flower. The highest carotenoid content of 549.68 µg/mg dry weight (DW) was observed in choline chloride: glucose (3:1) combination. Process optimization using Taguchi's model and response surface methodology (Box-Behnken design) was further employed to identify the optimal process parameters, viz., ultrasonication time as 35 min, incubation temperature as 25 °C, and incubation time as 2 h; a significant increase in carotenoid yield of 971.31 µg/mg DW of floral biomass was observed. Stability studies showed that carotenoids were stable in the dark and at a storage temperature of −20 °C. High performance liquid chromatography analysis revealed the presence of lutein as a major compound in the selected NADES system. Lastly, food safety studies of marigold floral NADES-based extracts were investigated. Antimicrobial and antioxidant activity studies of selected extracts revealed optimal properties indicating their potential to be formulated into food colorants and nutraceuticals.

1. Introduction

Global food safety report in 2024 that one in ten people would be ill every year because of contaminated food (<https://www.fao.org/food-safety/en/>). Synthetic dyes or artificial colorants commonly used as food additives pose several health problems including autism spectrum disorder (ASD) and attention deficit hyperactivity disorder (ADHD) in pediatric population, neurotoxicity, adult obesity, and other disorders (Amchova, Kotolova, & Ruda-Kucerova, 2015; de Oliveira et al., 2024). These trends in food safety showcase the need for developing safer and enriched functional food products. With increasing awareness of sustainable development goals, the need to develop technologies which are environmentally friendly, cost effective and involve waste valorization has also increased (<https://sdgs.un.org/goals>). Safe alternatives for synthetic dyes could be pigments from natural viz. flowers, fruits and vegetables having a high content of anthocyanins (red to blue),

carotenoids (yellow to orange) and betalains (red and yellow) (Ghosh et al., 2022). However, these pigments can degrade faster in adverse environment. (Ghosh et al., 2022). Amongst the pigments, carotenoids also have several health benefits including antioxidant activity, preventive properties against cancer, cardiovascular diseases, and eye health disorders (Cazzonelli, 2011).

Marigold (*Tagetes erecta* L.) flowers are often yellow or orange in color (Rodrigues et al., 2019). These flowers are rich in carotenoids including β-carotene and lutein (Rodrigues et al., 2019). Along with its health benefits, marigold loose flowers are widely used for decoration, traditional and cultural practices in India. The floriculture sector in India involves approximately 255,020 ha of marigold flowers and around 1,754,500 tons of flowers cultivated annually (Kaur et al., 2022). Post-harvest floral waste from religious and commercial activities and unsold flowers, as well as remains of unutilized flowers, contribute to environmental concerns (Waghmode, Gunjal, Nawani, & Patil, 2018).

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Thus, the valorization of marigold waste biomass through efficient and effective extraction techniques can serve as a sustainable approach for the recovery of bioactive compounds. Traditional extraction methods rely on organic solvents such as hexane, ethanol, and acetone (Amado et al., 2014). Even though they are effective, these solvents pose environmental and health risks, require high energy inputs, and often result in low extraction efficiency and thermal degradation (Sportiello et al., 2025). As a sustainable alternative, green extraction techniques like the use of natural deep eutectic solvents (NADES) have gained attention, focusing on reducing solvent toxicity and waste production. NADES are biodegradable, non-toxic solvents formed by mixing two or more bio-based compounds (hydrogen bond donors and hydrogen bond acceptors), with respect to solvent properties for extraction of both polar and non-polar biomolecules (Boudesocque-Delays et al., 2024). Compared to conventional organic solvents, NADES provides higher solubility, enhanced stability, and better bioavailability of target molecules while significantly reducing environmental impact.

The non-polar nature of carotenoids has led to the use and development of different hydrophobic NADES based extraction from sources like pumpkin, microalgae, and persimmon peels (Al Fuhaid et al., 2024; Plaza & Marina, 2025). Solvent systems containing menthol, thymol, and fatty acid esters including capric acid, caprylic acid, lauric acid, have been developed for extraction of carotenes having long chains of hydrocarbons amongst the carotenoid family of pigments (Osamede Airouyuwa et al., 2024). Relatively few studies on extraction xanthophylls, especially lutein (oxygenated form of carotene) from various natural sources including marigold flowers have been taken up. A study by Stupar et al. (2021) showed switching polarity of the NADES based system could recover carotenes from pumpkin tissue. Optimization of NADES-based extraction involves tuning key parameters such as solvent composition, temperature, extraction time, and solid-to-liquid ratio to maximize the yield of bioactive compounds. Advanced modelling techniques, including response surface methodology (RSM) and artificial neural networks (ANN), are used to predict optimal conditions and enhance extraction efficiency (Guddi & Sarkar, 2024).

The main goal of this study was to optimize a natural deep eutectic solvent-based extraction technology from used marigold flowers. After a thorough screening of HBD-HBA Extraction conditions were further optimized using Taguchi and response surface methodology (Box-Behnken design) techniques. Lastly, under optimal extraction conditions, preliminary food safety studies (microbial and heavy metal load), antioxidant and antimicrobial properties of NADES-based carotenoid extracts were evaluated for future food fortification studies.

2. Materials and methods

2.1. Plant material and chemicals

The marigold flowers were collected locally from Pune, Maharashtra. The drying of flowers was done using a hot air oven (Biotechnics India) at 50 °C (Siriamornpun et al., 2012). Tartaric acid, glucose, ethanol, and choline chloride (AR grade; >98 % pure) were purchased from SRL India Pvt Ltd. HPLC-grade methanol and methyl tertiary butyl ether, were purchased from Merck, India. Analytical standards β -carotene was purchased from Hi-Media India and Lutein was purchased from Sigma Merck, India.

2.2. Preparation of NADES

Different combinations of NADES as enlisted in (Table 1) using glucose and tartaric acid as hydrogen bond donors (HBD) and choline chloride as hydrogen bond acceptors (HBA) were prepared using the heating and stirring method described by Wu et al. (2022) with slight modifications. The water percentage of all the NADES systems was kept constant at 30 % (v/v). Post-preparation of NADES systems, dilution with ethanol (10 %; v/v) was carried out.

Table 1

Preliminary screening of ten combinations of NADES for carotenoid content.

NADES (code)	HBA: HBD	Molar ratio	Appearance	Total carotenoid content ($\mu\text{g}/\text{mg}$ dry weight)
NADES (N1)	Choline chloride: Glucose	1:3	Transparent	288.08 $\pm$ 4.34 ^a
NADES (N2)	Choline chloride: Glucose	1:1	Milky (Solid)*	NA
NADES (N3)	Choline chloride: Glucose	2:1	Transparent	340.78 $\pm$ 8.23 ^b
NADES (N4)	Choline chloride: Glucose	1:2	Milky (Solid)*	NA
NADES (N5)	Choline chloride: Glucose	3:1	Transparent	649.68 $\pm$ 3.84 ^c
NADES (N6)	Choline chloride: Tartaric acid	1:3	Transparent	490.17 $\pm$ 16.05 ^d
NADES (N7)	Choline chloride: Tartaric acid	1:1	Transparent	183.34 $\pm$ 0.38 ^e
NADES (N8)	Choline chloride: Tartaric acid	2:1	Transparent	214.00 $\pm$ 6.84 ^e
NADES (N9)	Choline chloride: Tartaric acid	1:2	Transparent	199.81 $\pm$ 0.86 ^e
NADES (N10)	Choline chloride: Tartaric acid	3:1	Transparent	236.85 $\pm$ 3.69 ^f

Note: All values are mean $\pm$ standard deviation of 3 replicates. Similar letters are not significantly different to each other at $p < 0.05$ by Duncan's multiple range test. *- Not fit for analysis.

2.3. Ultrasound-assisted extraction of carotenoids using NADES and estimation of total carotenoid content

Dried marigold petals (50 mg) were macerated and extracted in two ml of NADES solvent having 10 % ethanol. The mixture was sonicated in a Labman LMUC-3 digital ultrasonic bath (40 kHz, 100W, Normal/Soft mode) for 15 min and then pelleted by centrifugation at 8000 rpm for 10 min (Núñez et al., 2024). The supernatant was separated and total carotenoid content was determined using the Shimadzu (UV-1900i) spectrophotometer by measuring absorbance at 450 nm (Kusumiyati et al., 2025). All experiments were carried out in triplicate. Total carotenoid content was estimated in relative amounts of β -carotene ($\mu\text{g}/\text{mg}$ dry weight (DW)). From the initial solvents screened (Table 1), the effect of sugars (glucose, fructose, and sucrose) in combination with choline chloride as NADES constituents having the same water content and post dilution with ethanol was also tested for maximal yield of carotenoids.

2.4. Physicochemical properties of NADES

Physico-chemical properties such as density, viscosity, polarity, and pH were measured for selected NADES systems. pH was measured using a digital pH meter (Contech, CPH-102). Density was estimated using the stopper bottle method and was recorded for all NADES systems in cm^3 (Paiva et al., 2014). Polarity was assessed using Nile Red, a solvatochromic dye sensitive to solvent polarity. Maximum absorbance was measured using a Shimadzu UV-Visible Spectrophotometer, and E_{NR} value was estimated using a formula adopted from (Viñas-Ospino et al., 2023). Viscosity (mm^2/s) was measured using an Ostwald viscometer (Smith et al., 2014). FTIR analysis was performed to identify functional groups in choline chloride, glucose, the selected optimal NADES solvent,

and its marigold floral extract. Spectra were recorded in the range of 4000–400 cm^{-1} with a resolution of 4 cm^{-1} using FT-IR Spectrophotometer (IR Sprit, Shimadzu).

2.5. Optimization of extraction conditions using Taguchi and response surface methodology

Process optimization of NADES-based carotenoid extraction involved systematic experimental design to enhance efficiency. Initially, the Taguchi design of experiments was employed, focusing on four critical parameters: ultrasonication time, incubation temperature, incubation time, and water content. This approach facilitated the identification of parameters significantly influencing the extraction process. Taguchi Data was analyzed using Microsoft excel. (https://www.ee.iitb.ac.in/~apte/CV_PRA_TAGUCHI.htm). Parameters showing significant effect on yield were selected for RSM. A Box-Behnken design incorporating three parameters (ultrasonication time, incubation temperature, and incubation time) at three levels was utilized to optimize extraction process with triplicates of the central level to evaluate the pure error. After the RSM was conducted, the model was fitted using the RSM package (Lenth, 2009) in RStudio Posit team, 2025, to determine the interaction and effect of three factors on the yield and find the optimal values of the three factors for better yield (<https://github.com/Amishagarg09/Box-Behnken-/>). A quadratic model based on the equation below was employed in the investigation of the response:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_{12} X_1 X_2 + \beta_{23} X_2 X_3 + \beta_{13} X_1 X_3 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \beta_{33} X_3^2 \quad (\text{eq.3})$$

where Y is the total carotenoid content in $\mu\text{g}/\text{mg}$ dry weight; β_0 is the intercept (the value of the fixed response at the center point of the design), β_1 , β_2 , and β_3 are the sonication time, temperature and incubation time, for linear, quadratic and interaction regression coefficients, respectively; X_1 , X_2 and X_3 are the independent factors for sonication time (minutes), temperature (degree Celsius) and incubation time (hours), respectively. Based on the preliminary experiments the levels were chosen. The design factors X_1 (5 and 35), X_2 (10 and 40) & X_3 (0 and 2) are those with low levels of -1 and high levels of $+1$. The central values (0) determined for the experiments were 20 min, 25 °C and 1 h, respectively, for X_1 , X_2 , and X_3 . After fitting the model, a validation experiment was carried out in triplicates to compare the experimental yield with the theoretical carotenoid yield calculated from the quadratic equation. To assess the effect of the optimized parameters for carotenoid extraction using NADES, the effect size or hedges' g was calculated using cohensD package in Rstudio.

The molecule Lutein was docked against NADES solvent computationally using the software (BIOVIA Discovery Studio, 2022) <https://www.3ds.com/products/biovia/discovery-studio>. The 3D SDF files for optimized solvent and lutein were downloaded from PubChem. For the DFT calculations, Orca (version 6.1) was used for which the input files were generated using Avagadro software (Hanwell et al., 2012) and the output files were visualised using ChemCraft. All quantum chemical calculations were performed using ORCA version 6.1 (Neese, Wennmohs, Becker, & Riplinger, 2020). Molecular orbital and geometry visualizations were performed using Chemcraft (<https://www.chemcraftprog.com>).

2.6. Thermal, storage and luminance stability studies for NADES-based carotenoid extract

The thermal stability of selected NADES-based flower extracts was carried out by incubating at three different temperatures (40 °C, 60 °C, and 100 °C) on a Thermoblock (Eppendorf thermomixer C). Total carotenoid content was recorded at four different time intervals, viz. 0, 30, 60, and 120 min. For storage stability, extracts were incubated at three different storage temperatures (25 °C, 4 °C, and -20 °C) for 30

days, and total carotenoid content as described in earlier section 2.3 was recorded every six days. The effect of luminance on the stability of extracts was estimated by incubating extracts in two different conditions, light and dark, for 120 days. Total carotenoid content was monitored after every 15 days. All the storage experiments were conducted in three biological replicates. Post estimation of total carotenoid contents, half-life, degradation rate constant was determined according to first order rate kinetics (Kirca, Ozkan', & Cemeroglu, 2003).

2.7. Column chromatography and HPLC analysis

The NADES-based extract was subjected to chromatographic separation using a C18-packed solid phase extraction cartridge. The column was flushed with water to remove hydrophilic NADES components, and the recovery of NADES was calculated using the dry weight method. Fractions were systematically eluted using hexane as the mobile phase, ensuring efficient separation of the carotenoids. The collected fractions were filtered using a syringe filter (0.22 μm) were subjected to high-performance liquid chromatography (HPLC) to identify and quantify the individual components. HPLC analysis of the hexane fraction was carried out on a Waters 1525 binary HPLC equipped with a photodiode array detector and a Teknokroma 250 $\times$ 4.6 mm, 5 μm Vela C30 column. The mobile phase consists of a gradient solvent system where solvent A is 0.2 % ammonium acetate in 80 % methanol and solvent B is methyl tertiary butyl ether. The gradient elution used 100 % A for 15 min, a step to 95 % A and 5 % B, at 15 min, followed by a linear gradient to 80 % A and 20 % B by 27 min and achieved 100 % A by 28 min (Fraser et al., 2000). All the samples were injected in triplicate and monitored at 480 nm. Identification and quantification of lutein and β -carotene were done based on retention time and spectral data matching with respective analytical standards.

2.8. Food safety and toxicity studies

To assess food safety and toxicity of NADES-based marigold floral extracts, total bacterial and fungal count, and heavy metal residue analysis were carried out. The total bacterial and fungal count of the selected NADES-based marigold flower extract were assessed using the spread plate technique following the guidelines outlined in the *Manual on Microbiological Examination of Food and Water*, (2024). For bacterial enumeration, sterile agar plates using Nutrient Agar (NA) were prepared and incubated overnight at 37 °C. Serial dilutions of the test samples (10^0 , 10^{-2} , 10^{-4}) were prepared, spread onto the plates, and incubated at 37 °C for 72 h. Similarly, for fungal count, dilutions were plated onto Potato Dextrose Agar (PDA) and incubated at 25 °C for 72 h. The experiments were performed in triplicate and colony counts were determined using a colony counter (Cluster scientific 2023-11-478).

To check heavy metal analysis a total of 0.5 g of homogenized sample was taken in PTFE tubes, followed by 4 ml of ultrapure water, 3 ml nitric acid, and 0.5 ml of hydrochloric acid. The sample was kept for pre-digestion followed by digestion on Anton Paar Multiwave 7000 digester for 40 min at maximum temperature 180 °C. The blank method was made using the same dilutions without a sample. The digested sample was diluted to a volume of 50 ml, and 0.5 ml internal standards were added. Prepared samples were injected in the ICP-MS (Shimadzu-2030). All the samples were analyzed at Maarc Labs Private Limited, Pune, Maharashtra.

2.9. Antimicrobial activity

The antibacterial activity of the selected NADES-based marigold flower extract was evaluated using the well diffusion assay against *Escherichia coli* (MTCC723; Gram-negative bacterium). For the assay, four wells were created on Mueller-Hinton Agar (MHA), and 100 μL of the selected NADES, marigold extract, water (negative control), and Ampicillin (positive control) were added. The plates were incubated at

37 °C for 24 h and the assay was performed in triplicate. The antifungal activity of the selected NADES-based marigold flower extract was tested against *Candida albicans* using the well diffusion assay. For the assay, four wells were made on Potato Dextrose Agar (PDA), and 100 µL of the selected NADES, marigold extract, and water (negative control) were added. The plates were incubated at 28 °C, and the assay was performed in triplicate. Zone of inhibition post 24h was monitored to check antimicrobial activity.

2.10. Antioxidant activities

The free radical scavenging activity of NADES-based marigold extracts estimated by DPPH (2,2-Diphenyl-1-picrylhydrazyl radical) assay (Brand-Williams et al., 1995). IC₅₀ values (concentration at 50 % inhibition) was calculated by performing regression analysis of percentage inhibition and extract concentration. The reducing power of NADES-based marigold extracts was estimated by FRAP assay adopted from (Benzie & Strain, 1996) with modifications. FRAP value was calculated as mM FeSO₄ equivalent per ml of extract.

2.11. Statistical analysis

All experiments were conducted in triplicate to ensure statistical reliability and reproducibility of the results. Duncan's Multiple Range Test (DMRT) was done using SPSS Statistics (Version 17.0, IBM) to analyze the statistical significance among treatment means of the samples. All graphical representations and figures were made using Origin Pro 8.5 software. Additionally, the Response Surface Methodology (RSM) was employed using the RSM package and effect size calculation using cohensD package in R Studio for optimization and modeling of experimental data.

3. Results and discussion

3.1. NADES preparation

In this study 10 Mten molar combinations of NADES named N1 to N10 were prepared using choline chloride, glucose and tartaric acid prepared in a defined molar ratio specified in (Table 1) in addition with 10 % ethanol(v/v) for increased solubility. Carotenoids are hydrophobic compounds, and their solubility increases in less polar environments, as earlier reported in Ashenafi et al. (2023). Ethanol can alter the aqueous environment by lowering the dielectric constant and increasing its ability to dissolve nonpolar molecules like carotenoids (Mohammadi et al., 2020). Furthermore, ethanol can penetrate plant cell walls and membranes more effectively than water which leads to better release of plant carotenoids (Saini & Keum, 2018). Most of the prepared NADES exhibited stability and stayed liquid at ambient temperature without phase separation, which suggests that the components had strong hydrogen bond interactions. When the ratio of hydrogen bond donor (HBD) to hydrogen bond acceptor (HBA) is 1:1, the level of hydrogen bonding and charge delocalization achieves equilibrium in a stoichiometric ratio, causing it to form a crystalline structure (Aravena et al., 2023). This leads to the formation of a milky solid phase, which indicates partial dissolution or phase separation of the components. Two formulations, N2 and N4, were unsuitable for further analysis because of this phenomenon.

3.2. Ultrasound-assisted extraction of carotenoids

The efficiency of various combinations of NADES in extracting carotenoids from dried marigold floral tissue was determined through a screening process. The solvent system N5, which included glucose and choline chloride in a 3:1 M molar ratio, had the highest carotenoid extraction yield (549.68 ± 3.84 µg/mg dry weight (DW)) among the evaluated solvents (N1 to N10; Table 1). Strong hydrogen bond

interactions and the ideal polarity of N5 enabled improved breakdown and diffusion of carotenoids from the floral tissues. Further, the impact of replacing the sugar molecule of N5 with fructose and sucrose total carotenoid content (TCC) extracted from marigold floral biomass was evaluated (Fig. 1A). The carotenoid extraction yield is superior when observed with the glucose-based NADES compared to those formulated with fructose or sucrose (Fig. 1B).

To understand the reason behind extraction efficiency differences, physicochemical characterization was carried out. The three solvent systems (choline chloride and glucose; choline chloride and fructose; choline chloride and sucrose) had a pH of 6.02, 4.97, and 4.86 respectively after dilution with ethanol. pH of the solvent system has been shown to affect the yield of bioactive molecules and its stability in the solvent systems (Stupar et al., 2021). The weak acidic environment of choline chloride: glucose environment might have aided carotenoid extraction in comparison to other solvent systems. Another factor influencing carotenoid yield is the polarity of the solvent system. Glucose, a small aldohexose, forms strong and stable hydrogen bonds with choline chloride, resulting in a well-structured and least polar solvent system ideal for solubilizing carotenoids. The polarity of the other two sugar-choline chloride-based systems were higher than glucose-choline chloride-based NADES system (Fig. 1A). Thus, this could be one of the reasons why total carotenoid content was also higher in glucose-choline chloride-based NADES system considering the hydrophobic nature of carotenoids. High viscosity can affect the extraction of biomolecules by inhibiting solute mass transfer leading to longer extraction times (Hikmawanti et al., 2021). Amongst the tested combinations, glucose: choline chloride in comparison to the other two solvents had the lowest viscosity (Fig. 1A). In contrast, fructose, a ketohexose, displays altered hydrogen bonding geometry, while sucrose, being a disaccharide, introduces steric hindrance and increases viscosity, both of which could have been possible reasons for reduced extraction efficiency in these solvent systems (Paiva et al., 2014). In the case of density, all three solvent systems had density more than water. NADES systems having glucose-choline chloride and fructose-choline chloride had a very similar density viz. 1.21 g/cm³ and 1.29 g/cm³; NADES system having sucrose as the sugar component had a slightly higher density of 1.31 g/cm³. Higher density often has been shown to reduce extraction efficiency due to strong intermolecular packing (Stupar et al., 2021). Thus, through this screening of NADES systems, glucose: choline chloride in a molar ratio of 3:1 was found to be optimal for the extraction of carotenoids.

To understand the hydrogen bond interactions in the selected NADES systems FTIR analysis of individual components, solvent with ethanol and extract was performed. The FTIR spectra reveals distinct functional group variations among ChCl, glucose, ethanol, N5 solvent, and NADES (N5)-based marigold extract. ChCl shows characteristic peaks at ~3200–3500 cm⁻¹ (O–H stretching), ~1400–1500 cm⁻¹ (C–N and CH₂ bending), and ~1000–1100 cm⁻¹ (C–O stretching), consistent with its ionic structure (Fig. 1C). Glucose displays strong O–H stretching and C–O vibrational peaks between 1000 and 1200 cm⁻¹, while ethanol exhibits broad O–H (~3300 cm⁻¹), C–H (~2900 cm⁻¹), and C–O (~1050–1150 cm⁻¹) stretches. The N5 solvent demonstrates broadened and shifted peaks, especially in the 3200–3500 cm⁻¹ and 1000–1700 cm⁻¹ regions, indicating extensive hydrogen bonding between ChCl and the hydrogen bond donor component (glucose or ethanol), characteristic of DES formation (Abbott et al., 2003; Dai et al., 2013; Smith et al., 2014). The extract spectrum shows more complex features, including pronounced peaks at ~3300 cm⁻¹ (O–H/N–H), ~1600 cm⁻¹ (aromatic C=C), and ~1000–1200 cm⁻¹ (C–O), signifying the presence of multiple bioactive functional groups. These new peaks in the extract, absent in the N5 solvent, suggest effective solubilization and extraction of carotenoids. The shifts and broadenings confirm non-covalent interactions between solvent and solutes, indicating extraction efficiency, which aligns with previous studies on the capability of DESs in natural product extraction (Zainal-Abidin et al., 2017).

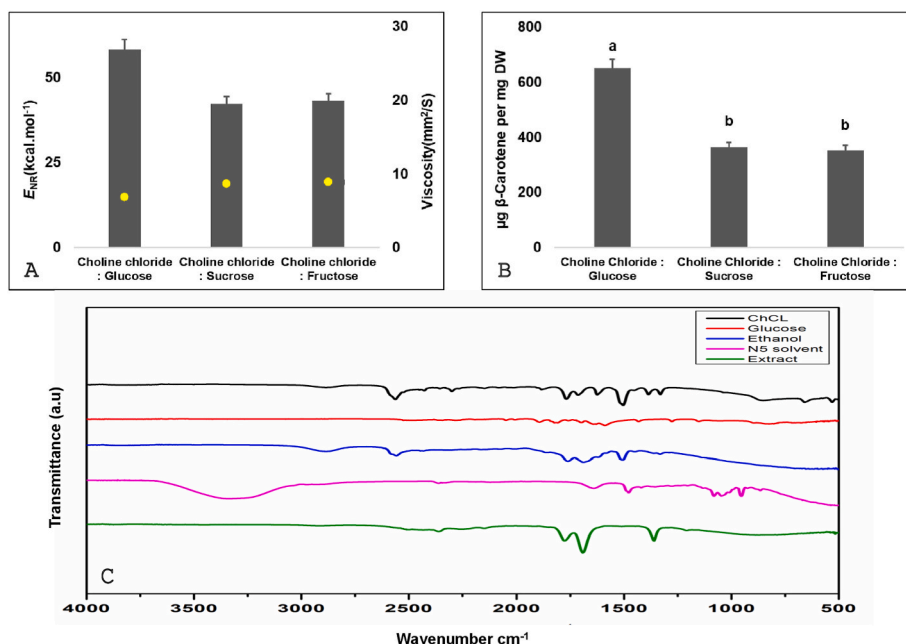


Fig. 1. (A) Polarity in terms of E_{NR} (kcal.mol⁻¹) and viscosity (mm²/s) of three NADES-based systems, (B) Total carotenoid content in µg/mg dry weight different NADES systems where sugar (glucose, fructose, and sucrose) as hydrogen bond donor with choline chloride as hydrogen bond acceptor, and (C) FTIR spectrum of choline chloride, ethanol, glucose, Choline chloride: glucose (3:1; N5 with ethanol) solvent, and its respective marigold floral extract. Note: Error bars represent the mean $\pm$ standard deviation of three replicates. Similar letters are not significantly different at $p < 0.05$ from each other by Duncan's multiple range test.

3.3. Carotenoid extraction process optimization using design of experiments

Taguchi design of optimization was carried out using L9 array as shown in [Supplementary Table S1](#). Four parameters namely water/moisture content, sonication time, incubation time and temperature were tested for carotenoid extraction using N5 solvent. Three levels of each parameter were tested. Based on the results obtained after analysis Temperature, incubation time and sonication time were selected for performing Box Benken RSM. The design for the experiment with response yield is shown in [Table 2](#). The model's efficacy was determined by the R^2 with value of 0.8848, which indicates that the quadratic model employed to infer the relationship between independent variables and dependent response is a good fit for the experiment. The p-value (0.06232) of the model fitted into the data suggests that the model is mildly significant with at least one parameter affecting the response. The F-statistic value of 4.267 suggests that some predictors contribute

significantly, and the level of significance ([Supplementary Table S2](#)) suggests that factor X2 with p-value 0.0328 has a significantly stronger effect as compared to X1 and X3.

Therefore, by employing coded variables for each parameter, a mathematical equation for total carotenoid content was formulated. This equation aids in predicting the theoretical yield for a set of parameters and also shows the magnitude of positive and negative impact on the yield. Equation (2) reveals that X3 has a major positive impact followed by X2, whereas, factors like X3², X1 and X1X3 have negative effects on the yield. The factors X2², X1², and X1X2 have a very insignificant impact on the total carotenoid content. The equation's intercept (600.05) makes it possible to predict the total carotenoid yield with accuracy. These results facilitate the formation of the second order polynomial equation that relates carotenoid yield (Y) to sonication time (X1), temperature (X2), and incubation time (X3).

$$Y = 600.05 - 10.56X_1 + 23.63X_2 + 130.51X_3 + 0.23X_1X_2 - 2.75X_1X_3 - 0.49X_2X_3 + 0.24X_1^2 - 0.62X_2^2 - 14.55X_3^2 \quad (\text{Equation 2})$$

Where, Y is the response variable (µg/mg), and X1, X2, and X3 are sonication time (minutes), temperature (°C), and incubation time (hours), respectively.

The surface plot ([Fig. 3](#)) shows that the carotenoid production first rises with sonication duration, but the yield either stabilizes or falls with increasing time. With an ideal range where yield is highest, temperature also has a significant effect and beyond this, carotenoids are likely degraded by high temperatures. A moderate temperature and sonication period are the best conditions for maximum yield, as shown by the ideal zone in contour plot ([Fig. 3D](#)). According to [Fig. 3B](#), the yield increases with incubation time, and the sonication period has a peak effect inferring that a short time may lead to incomplete extraction, while an excessively long time may cause carotenoids to decay. [Fig. 3E](#) contour plots show a wide range of carotenoid yield, indicating that sonication has a sharper threshold and incubation time has an increasingly enhancing impact. The saddle-like shape ([Fig. 3C](#)) indicates that temperature and incubation duration work together to increase the

Table 2

Box-Behnken design for optimizing carotenoid extraction from marigold flowers.

Sonication time (min)	Temperature (°C)	Incubation time (h)	Yield (µg/mg dry weight)
X1	X2	X3	
5	10	1	885.47 $\pm$ 64.0
5	40	1	614.77 $\pm$ 4.5
35	10	1	810.83 $\pm$ 8.91
35	40	1	754.10 $\pm$ 22.3
5	25	0	731.21 $\pm$ 12.0
5	25	2	947.66 $\pm$ 19.0
35	25	0	920.05 $\pm$ 29.9
35	25	2	971.30 $\pm$ 36.3
20	10	0	709.07 $\pm$ 29.0
20	10	2	725.74 $\pm$ 51.3
20	40	0	683.69 $\pm$ 19.1
20	40	2	670.50 $\pm$ 18.2
20	25	1	845.91 $\pm$ 23.9
20	25	1	850.63 $\pm$ 20.6
20	25	1	861.33 $\pm$ 39.4

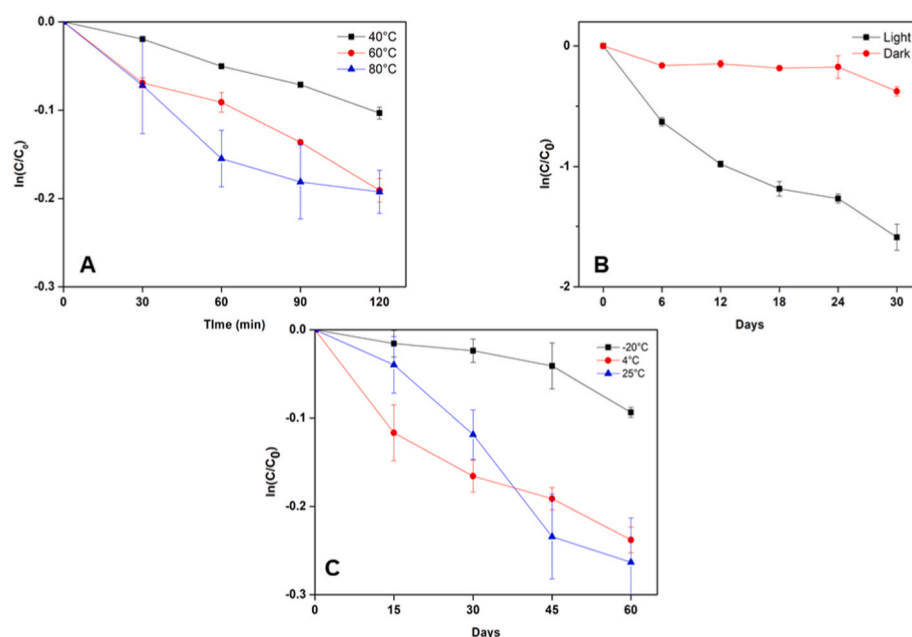


Fig. 2. Degradation rate curves showing stability of carotenoids (A) temperature ranges (40 °C, 60 °C, and 80 °C), (B) light and dark conditions for 30 days and (C) storage conditions in three different storage temperatures (−20 °C, 4 °C, and 25 °C) for 60 days.

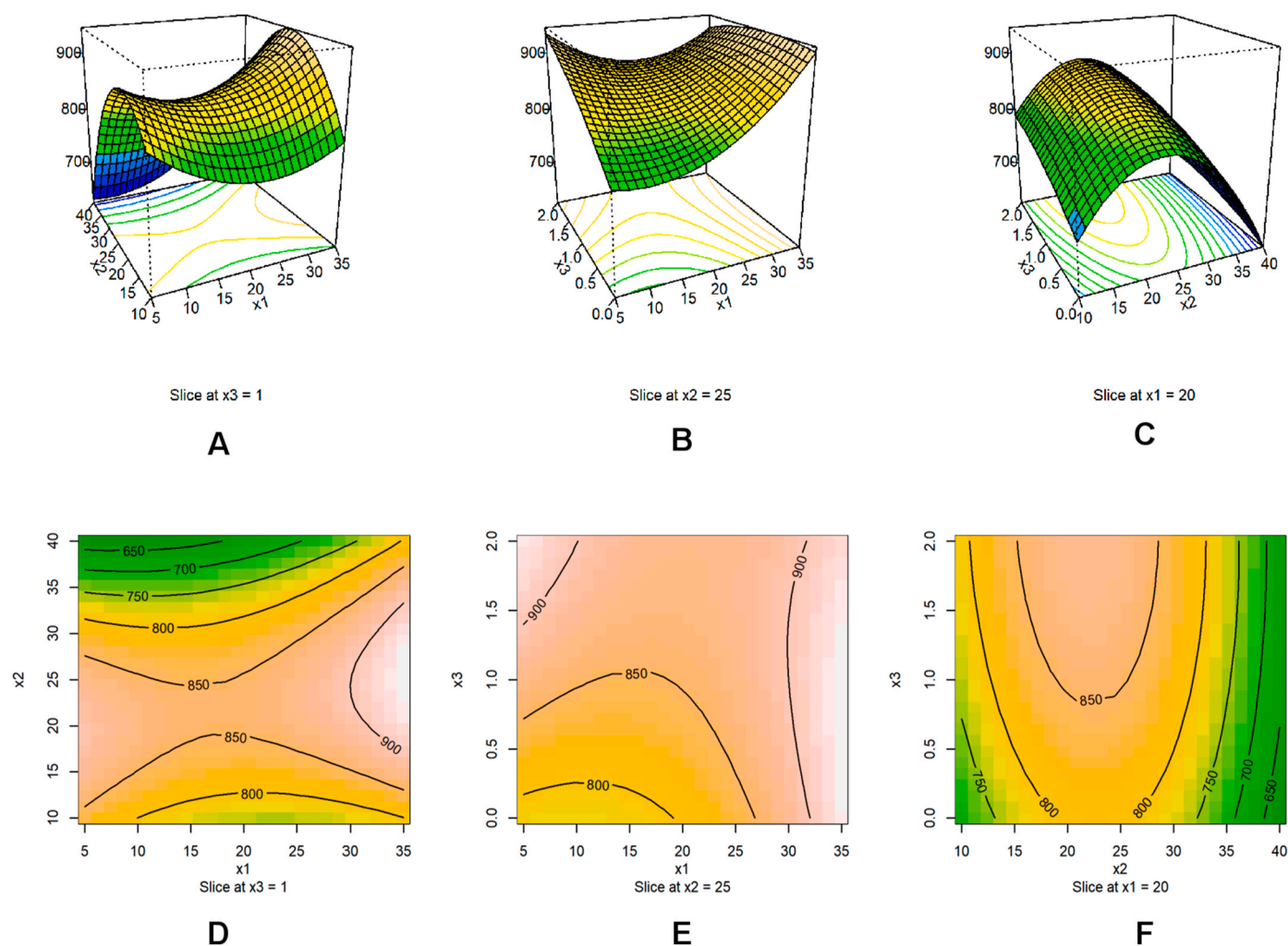


Fig. 3. Surface and contour plots (A) & (D) sonication time (X1) vs. temperature (X2); (B) & (E) incubation time (X3) vs. sonication time (X1); (C) & (F) temperature vs. incubation time.

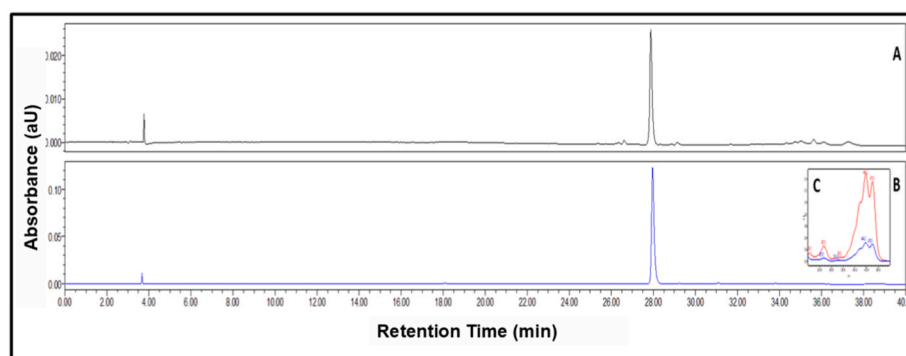


Fig. 4. HPLC Chromatogram of marigold NADES-based extracts (A) and Lutein standard (B) showing a match of retention time (27.67 min) and its corresponding UV-Visible spectral overlay (inset; C).

carotenoid yield and no factor alone can do so. Additionally, a particular temperature window where maximal yield occurs is indicated by contour plots (Fig. 3F). A set of stationary parameters were suggested after fitting the model into the data with the implication that those will provide the best carotenoid yield. The optimized extraction conditions of 22 min of ultrasonication, 22 °C incubation temperature, and a 2-h incubation time significantly increase the efficiency of carotenoid extraction from marigold (*Tagetes erecta*) petals when using natural deep eutectic solvents (NADES) (see Fig. 3).

To validate this, an experiment with test parameters (as suggested by the model) and control parameters (0 level parameters) (Table 3) was set up. The resultant yield showed a significant 34 % increase in the TCC yield. Equation (2) was used to get the theoretical yield (875.25 µg/mg dry weight) for the test parameters, which was then compared to the experimental yield to confirm that the model is functional. To assess the effect of optimized parameters, effect size or hedge's *g* was calculated. Hedges' *g* statistic expresses the difference between the two groups using the pooled standard deviation for a small sample size. In this study, the resultant hedges' *g* value was 7.92 (95 % CI: 5.49 to 10.35), which suggests the optimized conditions have a significantly large effect on the carotenoid yield as compared to the control ones. A *g* of 7.92 suggests that optimism is almost eight standard deviations higher than the control mean. These hydrophilic solvents create a highly nurturing microenvironment that provides the solubilization of carotenoids through strong hydrogen-bonding interactions.

Further to validate, the interactions between optimized solvent system and lutein, quantum chemical calculations was performed. Lutein, a long-chain carotenoid, interacts closely with N5 NADES solvent made of choline chloride, glucose, ethanol, and water. The central cluster of N5 forms many hydrogen bonds using -OH groups from the glucose, ethanol, and water, which interact with the terminal hydroxyl groups of lutein. (Fan et al., 2022). The choline parts of NADES can stick to electron electron-rich middle part of lutein while the chloride ions interact through light electrical forces. (Zhang et al., 2024). These forces make lutein dissolve in solvent even though lutein is not water soluble. The figure also shows lutein partly embedded inside the N5 NADES cluster, which supports findings from earlier research showing better

solubility and protection of lutein using natural solvents. The very negative total energy value (−2809.39 Eh) shows that the lutein–NADES N5 system is thermodynamically stable, meaning the molecules are strongly held together. (Supplementary Fig. S1).

3.4. Stability studies

The thermal stability of carotenoids in NADES-based extracts was evaluated at 40 °C, 60 °C, and 80 °C by calculating their half-life values, as shown in Supplementary Table S3. The degradation rate curves are presented in Fig. 2A. The results indicate that the most rapid degradation occurred at 80 °C, with a half-life of 433.21 min, followed by 60 °C with a half-life of 462.09 min, and the least degradation at 40 °C, with a significantly higher half-life of 770.16 min. These findings align with expected thermal degradation behavior, where increasing temperature accelerates carotenoid breakdown. It was earlier reported that carotenoids degrade faster at higher temperatures (Moura et al., 2023). The degradation of carotenoids in N5 extract varied significantly across all the storage temperatures as reflected in the half-life values. At −20 °C, degradation was minimal, with the highest half-life value (577.62 days) (Fig. 2B). This indicates that lower temperatures effectively preserve carotenoids by slowing the degradation process. At 4 °C and 25 °C, the half-life was shorter. These findings prove the importance of lower temperatures for preserving carotenoids in NADES-based extract, making −20 °C the most suitable condition for long-term preservation of carotenoids. In the case of photosensitivity/stability in light study, the degradation of carotenoids was much higher and faster in light exposure, as carotenoids are light-sensitive (Krinsky, 1989). Half-life values indicate a drastic change in the degradation rate in light (11.95 days) and dark (63.59 days) conditions, with half-life values of 11.95 and 63.59 days (Fig. 2C). Upon performing repeated experiments of stability (data not described in detail here) it was also noticed that quality of dried marigold floral biomass would also affect stability (half-life) and yield of carotenoids.

3.5. Column chromatography

Column chromatography was utilized to evaluate recovery of N5 solvent and the separation of carotenoids by switching solvents that reversibly change polarity (Jessop et al., 2010). Two-step column elution shows effective separation of NADES and carotenoids. First elution with water removed the NADES with 20 % column recovery. Similar study was carried out earlier where Recyclability of NADES was assessed by resin adsorption–desorption, achieving 99.5 % anthocyanin recovery and 96.8 % solvent recovery after dilution with 50 % water ((Panić et al., 2019)). The hydrophilic components of NADES, such as choline chloride and glucose, are highly soluble in water and are therefore eluted first during the initial water wash, while the hydrophobic carotenoids remain bound to the C18 column due to their strong

Table 3

Validation experiment yield with test and control parameters where X1 is sonication time (min), X2 is incubation temperature (°C) and X3 is incubation time (h).

	R1	R2	R3	Average Yield (µg/mg dry weight)
Test (X1=22; X2=22; X3= 2)	873.27	837.61	855.61	855.50
Control (X1=20; X2=25; X3=1)	667.77	618.34	625.22	637.11

Note: In table R1, R2 and R3 are three biological replicates.

affinity for the nonpolar stationary phase which was hexane (Bell et al., 1997). The FTIR spectra of N5 (black) and the eluate (red) show several overlapping peaks, indicating structural similarity between the original NADES solvent and the recovered eluate. Both spectra exhibit strong broad bands around $3300\text{--}3400\text{ cm}^{-1}$, corresponding to O–H stretching, suggesting preserved hydrogen bonding. Similar absorption peaks near 1600 cm^{-1} and $1000\text{--}1200\text{ cm}^{-1}$ indicate retained C=O and C–O functional groups, respectively (Supplementary Fig. S2).

3.6. HPLC-DAD analysis of NADES-based carotenoid extracts

HPLC analysis was conducted on the hexane fraction eluted from marigold flowers to investigate its phytochemical composition. The obtained chromatogram of sample and comparison with lutein standard has been showed in Fig. 4 (A and B) respectively. In the inset, comparison of UV-Visible spectra of both the standard and sample peaks are shown (Fig. 4C). The analysis revealed lutein as the predominant compound present in the marigold NADES-based extract, as previously reported in Rodrigues et al. (2019). The concentration of lutein in the hexane fraction was found to be $491.84\text{ }\mu\text{g/mL}$. In comparison, ultrasound-assisted extraction with green solvent like 2-methyltetrahydrofuran produced up to $41.5 \pm 1.4\text{ }\mu\text{g/mL}$ of lutein, previously reported in Rahman et al. (2025). Similarly, subcritical CO_2 enhanced with ultrasound-assisted achieved just 124.01 mg/100 g at optimized conditions (Fan et al., 2015). Enzyme-mediated processes, though industrially scalable operations, reported high operational costs despite $\sim 97\%$ carotenoid recovery (Barzana et al., 2002). In contrast, our NADES-based strategy not only achieves a markedly higher lutein concentration but also embraces green chemistry principles, offering a sustainable, non-toxic, and industrially viable alternative to traditional extraction technologies. This significant quantity highlights marigold flowers as a rich natural source of lutein (Alotaibi et al., 2021).

3.7. Food safety studies

3.7.1. Total microbial count

The NADES based extract showed with zero microbial count, as illustrated in Supplementary Fig. S3. The presence of choline chloride and glucose in NADES influence microbial growth, as it can utilize these components as a source of nutrition. Despite this the carotene-based extract showed no microbial count, suggesting its safety for applications in edible products, food fortification, nutraceuticals, and cosmetic formulations (Bedair et al., 2024).

3.7.2. Heavy metal residue analysis

The quantitative heavy metal residue profile of N5 extract was evaluated to test the safety of the extracts and compliance with regulatory standards. The maximum residue limits (MRL) stated by the Food Safety and Standards Authority of India (FSSAI) are lead-3 ppm, cadmium-1 ppm, mercury-3 ppm, and arsenic-1.8 ppm. The results demonstrate that the concentration of all the heavy metal residues is significantly below the maximum residue limit and was present in trace amounts ($<1\text{ ppm}$), confirming that N5 extract is safe for edible purposes. The N5 marigold extract, assured as a safe natural ingredient for human consumption, is further supported by its compliance with the FSSAI MRLs.

3.8. Bioactivity assessment of NADES based extract

The DPPH assay was used to estimate the antioxidant activity of NADES based marigold flower extract. The IC_{50} value of extract was determined to be $0.109 \pm 0.00443\text{ }\mu\text{g/mL}$ showcasing its effective free radical scavenging capacity. This moderate IC_{50} value suggest the presence of bioactive compounds with antioxidant potential. The ferric-reducing power was also evaluated for NADES-based extracts from marigold flowers. The reducing power as indicated by an FRAP value

7.04 mM FeSO_4 equivalent/mL of extract.

The NADES-based extract did not exhibit any antimicrobial activity with bacterial and fungal strains as shown in Supplementary Fig. S4. The pH remains within a range that supports bacterial and fungal growth and does not have an inhibitory effect. The microbial growth of the tested pathogens is due to the biocompatible nature of choline chloride and glucose, which may serve as a metabolic substrate rather than exhibiting no inhibitory effect on microbial growth (Radošević et al., 2018). However, the presence of microbial growth also highlights potential of NADES system for the application in formulation, such as probiotic preservation, enzyme stabilization, or as a medium bioactive compound extraction where antimicrobial activity is not the primary requirement.

4. Conclusion

Choline chloride-based NADES in a molar ratio of 3:1 yielded the highest carotenoid content from marigold floral biomass. Process optimization and modeling of carotenoid extraction significantly improved the extraction yield by 34% . HPLC analysis confirmed the presence of major carotenoids, such as lutein. Considering marigold flowers are a rich source of carotenoids, including lutein, the optimized NADES-based extracts can now be directly incorporated for the development of functional food products (nutraceuticals). Compared to conventional organic solvents like hexane or acetone, the NADES-based extraction method offers a greener, non-toxic, and biodegradable alternative with significantly higher carotenoid yield ($971.31\text{ }\mu\text{g/mg DW}$). It operates under mild conditions (25°C), preserving compound stability and bioactivity. Additionally, NADES extracts show better food safety and hence suitable for nutraceutical and food applications. Considering the fact that NADES based extraction technologies are greener and are also considered as GRAS by FDA, industrial scale up of optimized method shows potential. Toxicological evaluations of the NADES extracts might further confirm the food safety/drug safety levels of these solvent systems. Another disadvantage of the NADES systems is their high boiling points which complicate metabolite recovery, making downstream processing (recovery of phytochemical) more difficult than conventional systems. Future research should explore the bioavailability of NADES-based extracts (from marigold flowers) under in vitro and in vivo systems to validate their functional efficacy. Additionally, evaluating the application of these extracts in real food systems will help establish their commercial viability as natural additives or nutraceutical ingredients.

CRediT authorship contribution statement

Atharva Joshi: Writing – original draft, Methodology, Investigation, Formal analysis. **Aarya Kunnure:** Investigation, Formal analysis. **Abhishek Aghamkar:** Investigation, Formal analysis. **Amisha Garg:** Writing – original draft, Software, Methodology, Investigation. **Tejaswini Pachpor:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis. **Nithya N. Kutty:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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analysis.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.lwt.2025.118549>.

Data availability

Data will be made available on request.

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